

The Standard Model of Particle Physics

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Motivation and Postulates

0.1 Motivation

Outline of Notes:

These notes follow from the lectures of Dr. Ruth Britto's lectures which are based on books by Peskin and Thompson. The Standard Model of Particle Physics is an attempt to answer the following questions:

- Why do some nuclei emit high energy particles?
- What specific reactions are responsible?
- What, in spite of the nucleus being comprised of positive protons and neutral neutrons, holds the nucleus together?
- What are protons and neutrons made of?

0.2 The Postulates

The Postulates of the Standard Model of Particle Physics are those of Quantum Field Theory namely the union of those of special relativity and non-relativistic quantum mechanics. Each of these formalisms rely on the axioms of the mathematical structure in which they are built.

0.2.1 The Postulates of Quantum Mechanics

- The quantum state of a mechanical system at any given time t_0 is represented by a vector in a Hilbert space $|\psi\rangle \in \mathcal{H}$. Vector spaces are closed under vector addition. Thus, for arbitrary vectors we have the principle of superposition $a|\psi\rangle + b|\xi\rangle \in \mathcal{H}$.
- The real coordinates z^i are represented at a given time t_0 by Hermitian operators $\hat{Z}^i$ which satisfy the following commutation relations

$$\begin{aligned} \frac{i}{\hbar} [\hat{P}_a^\alpha, \hat{X}_b^\beta] &= \delta_{ab} \delta^{\alpha\beta} \hat{I}, & [\hat{X}_a^\alpha, \hat{X}_b^\beta] &= 0, & [\hat{P}_a^\alpha, \hat{P}_b^\beta] &= 0, & \alpha, \beta &= 1, 2, 3 & a, b &= 1, \dots, N \\ \frac{i}{\hbar} [\hat{S}_a^\alpha, \hat{S}_b^\beta] &= -\delta_{ab} \varepsilon^{\alpha\beta\gamma} \hat{S}_b^\gamma, & \sum_\alpha (\hat{S}_a^\alpha)^2 &= s_a(s_a + 1) \hbar^2 \hat{I}, & [\hat{S}_a^\alpha, \hat{X}_b^\beta] &= 0, & [\hat{S}_a^\alpha, \hat{P}_b^\beta] &= 0 \end{aligned}$$

where s_a is the spin of the a -th particle.

XXX Finish later XXX

1 Symmetries of Spacetime

1.1 Relativistic Particle Kinematics

An isolated particle has an energy-momentum four-vector

$$p^\mu = (E, \vec{p}c)^\mu. \quad (1.1)$$

Under a boost by v in the $\hat{3}$ direction:

$$E' = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \left(E + p^3 \left(\frac{v}{c} \right) c \right) \quad p^{3'}c = \gamma \left(p^3 c + \frac{v}{c} E \right) \quad (1.2)$$

$$p^{1,2'} = p^{1,2}. \quad (1.3)$$

which may be written in matrix form as $p' = \Lambda p$ with

$$\Lambda = \begin{bmatrix} \gamma & 0 & 0 & \gamma\beta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \gamma\beta & 0 & 0 & \gamma \end{bmatrix} \quad \beta = \frac{v}{c} \quad \gamma = \frac{1}{\sqrt{1 - \beta^2}}. \quad (1.4)$$

In Einstein's notation, $p^{\mu'} = \Lambda^{\mu'}_{\nu} p^\nu$. We will work in the mostly negative convention for the metric. Thus, $p \cdot q = E_p E_q - \vec{p} \cdot \vec{q}$

$$\eta_{\mu\nu} = \eta^{\mu\nu} = \text{diag}(+, -, -, -) \quad (1.5)$$

Thus we define the inner product

$$p \cdot q = p^\mu \eta_{\mu\nu} q^\nu \quad q_\mu = \eta_{\mu\nu} q^\nu = (E_q, -\vec{q})_\mu \quad (1.6)$$

A useful Lorentz invariant (quantity that is independent of the motion of the particle, thus necessarily a scalar) is

$$p_\mu p^\mu = E^2 - |\vec{p}|^2 c^2 = m^2 c^4 \quad (1.7)$$

In the rest frame

$$p^\mu = (E_0, \vec{0})^\mu \quad E_0 = mc^2 \quad (1.8)$$

$$\implies (mc^2)^2 = p^2 = E^2 - |\vec{p}|^2 c^2 \quad (1.9)$$

$\exists$ two equivalent notations for p^2

$$p^\mu = (E_p, \vec{p}c)^\mu \quad p^\mu = mc^2 \gamma (1, \vec{\beta})^\mu \quad (1.10)$$

1.2 Natural Units

In natural units, we set

$$c = \hbar = \mu_0 = \epsilon_0 = 1 \quad \implies \quad h = 2\pi. \quad (1.11)$$

Note that momentum and mass have units of Energy, and Length and Time have units of Inverse Energy. The Coloumb potential is

$$V(r) = \frac{Q_1 Q_2 e^2}{4\pi\epsilon_0 r} \quad (1.12)$$

We can identify a dimensionless quantity

$$\alpha = \frac{e^2}{4\pi\epsilon_0 \hbar c} \approx \frac{1}{137} \quad (1.13)$$

This coupling constant is called the fine structure constant, and it is relatively small. We conclude that the electromagnetic interaction is a weak interaction.

We naturally ask, *Can the strong and weak interaction also be characterised by a dimensionless coupling constant?*

They can! Both the strong and the weak interaction are also weak. Strong > weak > electromagnetism.

1.3 Theory of Discrete Groups

In quantum mechanics groups are discussed on two levels:

- **Abstract:** A group is a set of elements $\mathcal{G} = \{a, b, \dots\}$ with a multiplication law defined $ab \in \mathcal{G}$ and obeys:
 Associativity $a(bc) = (ab)c$.
 Identity: $\exists 1 \in \mathcal{G}$ such that $1a = a1 = a$.
 Inverse: $\forall a \in \mathcal{G}, \exists a^{-1}$ such that $aa^{-1} = a^{-1}a = 1$
- Abstract operators act on the space of states. A group transformation is a symmetry of a quantum system if it leaves the dynamics invariant. This is true if the transformations commute with the Hamiltonian.

In quantum mechanics states lie in a Hilbert space, $|\psi\rangle \in \mathcal{H}$. Symmetries convert these into another, (an automorphism conserving norms).

In quantum mechanics a group $\mathcal{G}$ is described by a set of unitary transformations $\{\mathcal{U}(a)\}$. If

$$ab = c \quad \implies \mathcal{U}(a)\mathcal{U}(b) = \mathcal{U}(c). \quad (1.14)$$

This set is called a unitary representation. The group $\mathcal{G}$ is a symmetry of a quantum mechanical system if $\forall a \in \mathcal{G} [\mathcal{U}(a), H] = 0$. Thus, if $|\psi\rangle$ is an eigenstate of H with energy E , $\mathcal{U}(a)|\psi\rangle$ will also be an eigenstate of H with the same energy. Given a group $\mathcal{G}$ with elements $\{a\}$, one can find a set of finite-dimensional unitary matrices that obey $\mathcal{U}(a)\mathcal{U}(b) = \mathcal{U}(c)$?

Yes, there are many examples e.g. the monster group, etcetera.

Example:

Take the group $Z_2 = \{1, -1\}$, satisfying $1 \cdot 1 = (-1) \cdot (-1)$ and $1(-1) = (-1)1 = -1$. One can represent Z_2 by 2×2 matrices. Consider two particles π^+ and π^- . Define the charge operator C

$$C|\pi^+\rangle = |\pi^-\rangle \quad C|\pi^-\rangle = |\pi^+\rangle. \quad (1.15)$$

The action of C on a 2-dimensional space

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ on } \begin{bmatrix} |\pi^+\rangle \\ |\pi^-\rangle \end{bmatrix}.$$

If $[C, H] = 0$ then the masses and decay rates of π^+ and π^- are the same. One can define the trivial operation

$$1|\pi^+\rangle = |\pi^+\rangle \quad 1|\pi^-\rangle = |\pi^-\rangle \quad (1.16)$$

which is given by the action of $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ on $\begin{bmatrix} |\pi^+\rangle \\ |\pi^-\rangle \end{bmatrix}$. As such, we have that the unitary matrices $\{1, C\}$ form a 2 dimensional unitary representation of Z_2 . If the matrices that comprise a unitary representation of a group $\mathcal{G}$ all commute with H then we say that H has a $\mathcal{G}$ symmetry.

A group $\mathcal{G}$ is abelian if

$$\forall \{a, b\} \in \mathcal{G}, \quad ab = ba.$$

A unitary representation of an abelian group is a set of abelian unitary matrices. This implies that they are simultaneously diagonalisable. If we call the eigenvalue c :

$$\begin{aligned} c = +1 : & \quad [|\pi^+\rangle + |\pi^-\rangle] \setminus \sqrt{2} \\ c = -1 : & \quad [|\pi^+\rangle - |\pi^-\rangle] \setminus \sqrt{2}. \end{aligned}$$

$C^2 = 1, \implies c = \pm 1$. There are two types of physical symmetries. Symmetries involving spacetime coordinates are called *space time symmetries*. Symmetries relating particles or quantum states are called *internal symmetries*.

If $\mathcal{G}$ is a non-abelian group, then, in general, one can't diagonalise simultaneously. However, by a change of basis we can write in block form

$$U_R \rightarrow \begin{bmatrix} U_1 & 0 & 0 \\ 0 & U_2 & 0 \\ 0 & 0 & U_3 \end{bmatrix}, .$$

where the U_i 's are as small as possible. The minimal size unitary transformations representing $\mathcal{G}$ are called irreducible unitary representations of $\mathcal{G}$. The size of the matrix U_i is called the *dimension* d_i of the representation.

Fact: $\forall$ discrete groups $\mathcal{G}$ with a finite number of elements, $\exists$ a finite number of inequivalent finite dimensional irreducible representations.

For any discrete group $\mathcal{G}$ with n elements the inequivalent unitary representations satisfy $\sum_i d_i = n$. Once we have the irreducible representations $U_i, \forall i$ then we have determined the structure of U_R .

1.4 Theory of Continuous Groups

There exists a continuous set of elements.

Examples include Spatial translations and Lorentze transformations.

A translation by a in 1 dimension

$$U(a) = e^{-iaP} \quad (1.17)$$

where P is the operator measuring the total momentum.

$$\langle x|\rho\rangle = e^{i\rho x} \quad \langle x|U(a)|\rho\rangle = e^{i\rho(x-a)} \quad (1.18)$$

The set of unitary operators is a unitary representation of the spatial translations. Since $U(a)$ is an exponential $\exists$ a relationship between the group of translations and the Hermitian operator P .

$$P \text{ is hermitian} \iff U(a) \text{ is unitary.} \quad (1.19)$$

$$U(a)^\dagger = e^{iaP^\dagger} = e^{iaP} = U(a)^{-1} \quad (1.20)$$

Hermitian operators $\iff$ observables. Conserved quantity $\iff [Q, H] = 0$. A non-abelian cont. group in 3d (e.g. the rotation group). $U(\vec{\alpha}) = e^{-i\vec{\alpha}\cdot\vec{J}}$, where $\vec{\alpha}$ give the direction and angle of the rotation and $\vec{J}$ are the operators of angular momentum. These operators obey the algebra

$$[J^i, J^j] = \epsilon^{ijk} J_k \quad (1.21)$$

If J^i satisfies (1.21) $\implies U(\vec{\beta})U(\vec{\alpha}) = U(\vec{\gamma})$. $\vec{\gamma}$ is a result of rotating through $\vec{\alpha}$ then $\vec{\beta}$.

The quantum states of atoms are organised into multiplets of atoms of definite angular momentum e.g $2D$ or $2P$ of hydrogen. States of a definite angular momentum give a finite-dimensional irreducible representation of the rotation group. The simplest rep of the rotation group

$$\begin{aligned} 1d : \quad J^i &= 0 \\ 2d : J^1 &= \frac{\sigma^1}{2} \quad J^2 = \frac{\sigma^2}{2} \quad J^3 = \frac{\sigma^3}{2} \\ \sigma^1 &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \sigma^2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad \sigma^3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \\ 3d : J^1 &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{bmatrix} \quad J^2 = \begin{bmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{bmatrix} \quad J^3 = \begin{bmatrix} 0 & -i & 0 \\ -i & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$

These reps give spin 0, $\frac{1}{2}$ and 1.

$\forall z \in \mathbb{Z}$ or $j \in \frac{\mathbb{Z}}{2}$, $\exists$ a set of $3(2j+1)(2j+1)$ matrices satisfying these commutation relations. *Spin j rep of rot group.*

How might one decompose the set of quantum states into irreducible representations of the rotation group?

Atoms have

- Orbital angular momentum ℓ

- Spin angular momentum s
- Total angular momentum j

Given a set of states with $(2\ell + 1)(2\ell + 1)$ elements. Then the total angular momentum j takes values $|\ell - s| \leq j \leq \ell + s$. Since $[\vec{J}, \vec{H}] = 0, \forall j, \exists$ a set of $(2j + 1)$ states with the same Energy. XXX See 4.1. XXX

Concider 3d rotations as an abstract group $SO(3)$ in general $SO(d)$. $d = 2$ is the case of rotations about a circle. An abelian group of translations with $\phi \sim \phi + 2\pi$.

$$e^{-i\phi} \rightarrow e^{-i\alpha} e^{-i\phi} \quad (1.22)$$

transformation by 1×1 matrices $U(1)$.

$U(n)$ is given by

$$U(n) = e^{-i\alpha^a t_a} \quad a = 1, \dots, n \quad t^0 = 1 \quad (1.23)$$

If we omit $t^0 = 1$ we get a non-abelian group with $n^2 - 1$ generators and traceless $SU(n)$. For $n \geq 2$ the Pauli matrices form a basis for the 2×2 matrices.

$$\implies SO(3) \equiv SU(2) \quad (1.24)$$

Lie group : A continuous group of transformations generated by hermitian matrices $U = e^{-i\alpha^a t^a}$ is called a lie group. The commutation algebra of the generators is the lie algebra of the group

$$[t^a, t^b] = i f^{abc} t_c \quad (1.25)$$

Fact: one can choose a basis for t^a such that the structure constants for f^{abc} are antisymmetric in $[abc]$. For the case of the rotation group $f^{abc} = \epsilon^{abc}$. The groups

$$\begin{aligned} U(1) &\equiv SO(2) \\ SU(2) &\equiv SO(3). \end{aligned}$$

and $SU(3)$ are important for particle physics.

1.5 Discrete Spacetime Symmetries

Special relativity includes continous symmetries i.e. rotations and Lorentz transformations. It also contains two discrete spactime transformations, whihc leave $\eta_{\mu\nu} = \text{diag}(+, -, -, -)$ invariant. Nother's theorem implies that if we have conservation of energy-momentum then we have the invariance of the equations of motion under space-time translations. Likewise, if we have the conservation of angular momentum then we have invariance under rotation and boosts.

/ $\exists$ a fundamental principle requiring extra discrete spacetime symmetries of the Hamiltonian of a quantity associated with discrete symmetries, i.e. no conserved quantity. These have only been determined experimentally.

We have Parity P and time reversal T which aren't apart of the continuous Lorentz group.

$$P^2 = 1 \quad T^2 = 1 \quad (1.26)$$

They are quantum operators with eigenvalues not equal to 1. Continuous Lorentz invariance does not imply P and T are conserved in electromagnetism and atomic physics. P and T are conserved under the strong interaction. Both are violated by the weak interaction.

$$\text{Parity} \quad X^\mu (x^0, \vec{x})^\mu \rightarrow (x^0, -\vec{x})^\mu \quad (1.27)$$

Take a rotation matrix, e.g.

$$\Lambda = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1.28)$$

or any Lorentz transformation with $\det \Lambda = +1$. However, (1.27) is implemented by a matrix with $\det \Lambda = -1$, and thus can't be generated by a product of continuous rotations.

$$\text{Time Reversal} \quad X^\mu = (x^0, \vec{x})^\mu \rightarrow (-x^0, \vec{x})^\mu \quad (1.29)$$

It can't be continuously generated. In quantum mechanics $\exists$ an intrinsic parity. Under parity its state of momentum $\vec{k}$ can transform as

$$P|A(\vec{k})\rangle = +|A(\vec{k})\rangle \text{ or } -|A(\vec{k})\rangle \quad (1.30)$$

Intrinsic parity $+1$ or -1 .

$\exists$ intrinsic time reversal as well.

$\exists$ another discrete symmetry in QFT.

Given a particle $\exists$ an anti-particle with the same mass and opposite conserved charge.

Charge Conjugation C

$$C^2 = \pm 1 \quad \implies \quad \text{Eigenvalues are } \pm 1 \quad (1.31)$$

A theorem in QFT states that the combination of CPT must be a symmetry of all particle interactions. It is a separate conversation to consider the conservation of the individual P , C , T which will be relevant when we consider the weak interaction.

2 Relativistic Wave Equations

- Standard quantum mechanics (1 particle in 0+1 dimensions) isn't sufficient. Schrodinger's equation isn't Lorentz invariant. It isn't symmetric with respect to spacetime. In a relativistic theory, the wave-functions of quantum mechanical particles obey equations which are symmetric with respect to space time.
- Standard quantum mechanics only has 1 particle, but we want to consider creation/annihilation.

QFT solves these generalisations.

2.1 The Klein-Gordon Equation

A wave equation is invariant under a group of symmetries, if $\forall$ solutions, the symmetry transform of that solution is another solution of the wave equation.

For a *scalar field* the Lorentz transform of a waveform is the same waveform evaluated at Lorentz transformed points.

$$\phi(x^\nu) \rightarrow \phi'(x^\nu) = \phi\left((\Lambda_\nu^\mu)^{-1} x^\nu\right) \quad (2.1)$$

For a Lorentz invariant theory waves should have if $\phi(x^\nu)$ is a solution then $\phi'(x^\eta)$ should too. The simplest example satisfying this is the Klein-Gordon KG equation

$$\left(\frac{\partial^2}{\partial t^2} - \nabla^2 + m^2\right)\phi(x^\mu) = 0 \quad (2.2)$$

note in x^μ we have that t and $\vec{x}$ appear symmetrically.

The 4 gradient $\partial_\mu := \left(\frac{\partial}{\partial t}, \frac{\partial}{\partial \vec{x}}\right)_\mu$ transforms under lorentze transformations as a four vector with lower index. Thus

$$p \cdot \partial = \left(E \frac{\partial}{\partial t} + \vec{p} \cdot \vec{\nabla}\right) \quad \partial^2 = \left(\frac{\partial^2}{\partial t^2} - \nabla^2\right) \quad (2.3)$$

are lorentz invariant operators. In a covariant form

$$(\partial^2 + m^2)\phi(x^\mu) = 0 \quad (2.4)$$

which has solutions

$$\phi(x^\mu) = e^{-iEt + i\vec{p} \cdot \vec{x}} = e^{-ip_\mu x^\mu} \quad (2.5)$$

where $p^\mu := (E, \vec{p})^\mu$ is a four vector satisfying

$$E^2 - |\vec{p}|^2 = p^2 = m^2 \quad (2.6)$$

The lorentze invariant of the four vector product

$$p_\mu x^\mu = (\Lambda p)(\Lambda x) \text{ or } (\Lambda p)(x) = p(\Lambda^{-1}x) \quad (2.7)$$

The boost of ϕ is $\phi'(x^\mu) = e^{-ip\Lambda^{-1}x} = e^{-i\Lambda p x}$. The KG equation has solution with both positive and negative $E = \pm E_p$.

How can we explain the negative energy? QFT. We can use the action to describe the lorentze invariance of the KG equation.

$$S[x(t), \dot{x}(t)] = \int dt L(x, \dot{x}) \quad (2.8)$$

The principle that S is stationary with respect to all variations of the solution $x(t)$ yields the equation of motion of the system. If $x(t) \rightarrow x(t) + \delta(x(t))$, then one can write

$$\delta S[x(t), \dot{x}(t)] = \int dt \delta x(t) [\epsilon[x(t), \dot{x}(t), \ddot{x}(t)]] \quad (2.9)$$

Thus the equations of motion is the set of equations for which $\epsilon = 0$. To get relativistic equations of motion, require relative invariant action S . The action should be a function of $\phi(x)$. Don't just integrate over t , integrate over all of minkovski space d^4x .

$$S[\phi(x)] = \int d^4x \mathcal{L}[\phi, \partial_\mu \phi] \quad (2.10)$$

where we chose $\mathcal{L}$ such that it is relativistically invariant and thus S is also. Consider the Klein-Gordon Lagrangian density

$$\mathcal{L} = \frac{1}{2} [\partial^\mu \phi \partial_\mu \phi - m^2 \phi^2] \quad (2.11)$$

No open indices thus it is Lorentze invariant. The variation with respect to $\phi(x)$ is

$$\delta \mathcal{L} = \partial_\mu \delta \phi \partial^\mu \phi - m^2 \delta \phi \phi \quad (2.12)$$

$$\implies \delta S = \int d^4x \delta \phi (-\partial^2 - m^2) \phi(t) \quad (2.13)$$

A solution of the equation is a stationary point of $S(\phi)$. If S is invariant under Lorentz transformations then the transform ϕ' will be too. Thus if ϕ is then ϕ' is too.

2.2 Fields and Particles

The Klein-Gordon equation is the relativistic extension of the Schrodinger equation which describes a single non-interacting point particle. We want to describe a multi-particle theory. In this attempt, we write a Hamiltonian which results in the Klein-Gordon equation and then quantise. We aim to obtain a Lorentz invariant ground state g 'The vacuum state', and excited states with the energy momentum of particles of mass m .

In the solution to the Klein-Gordon equation the field $\phi(x)$ becomes an operator $\hat{\phi}(x^\mu)$ that can create or destroy particles.

Let $|0\rangle$ be the ground state of the Hamiltonian H_{KG} . This is a vacuum state and thus, contains no particles. Let $|\varphi(p)\rangle$ be an excited state with 2 particles of momentum p . This a state of higher energy, with energy E_p above the vacuum energy E .

The operator $\hat{\phi}(x^\mu)$ has a zero matrix element corresponding to the destruction of the particle

$$\langle 0 | \hat{\phi}(x^\mu) | \varphi(p) \rangle = e^{-ip_\mu x^\mu} \quad (2.14)$$

Since $\hat{\phi}(x^\mu)$ obeys the Klein–Gordon equation, then $e^{-ip_\mu x^\mu}$ is a solution also.

The p^0 component is the positive energy solution from $E_p = \pm E$.

The field operator $\hat{\phi}(x^\mu)$ destroys the particle at spacetime point x^μ .

As such, we can interpret the right–hand side of (2.14) as the wavefunction that the particle occupied at the event in which it was destroyed. Taking the conjugate gives

$$\left\langle \varphi(p) \left| \hat{\phi}^\dagger(x^\mu) \right| 0 \right\rangle = e^{ip_\mu x^\mu} \quad (2.15)$$

We see the negative energy solution of the Klein–Gordon equation on the right–hand side. This is natural since the particle φ appears in the bra vector, so the right–hand side would be the complex conjugate of the Schrödinger equation.

In general $\phi(x^\mu) \in \mathbb{C}$ so in general $\phi^\dagger \neq \phi$. Hence, we have a new matrix element when ϕ has an imaginary component

$$\left\langle 0 \left| \hat{\phi}(x^\mu) \right| \varphi'(p) \right\rangle = e^{-ip_\mu x^\mu} \quad (2.16)$$

where $\varphi'(p)$ is a new particle which is distinct from φ . Should $\varphi(p)$ have electric charge Q then $\varphi'(p)$ has electric charge $-Q$. The particles φ and φ' could be identical. In this case, the particle destroyed by $\phi(x^\mu)$ can be its own anti–particle. There exists various formulae for these cases which can be found in Peskin Appendix C. XXX

2.3 Maxwell’s Equations

The particle that appears earlier had no quantum number except for energy and momentum. We know that particles have intrinsic angular momentum, for example electrons have spin $\frac{\hbar}{2}$ and photons have spine $1 \cdot \hbar$.

Spin 1 is the vector representation of angular momentum (one open index).

Consider a 3–vector field $V^i(\vec{x})$, which transforms under rotations

$$V^i(\vec{x}) \rightarrow V'^i(\vec{x}) = R_{ij} V^j(R^{-1}\vec{x}) \quad (2.17)$$

A rotation moves the coordinates of the field in the same way as $\phi \rightarrow \phi' = \phi(\Lambda^{-1}x)$, but also changes the orientation of the field.

Definition 2.3.1 (Polarisation Vector). If $V^i(\vec{x})$ is an operator and destroys a particle v , the matrix element corresponding to that operation is

$$\left\langle 0 \left| V^i(\vec{x}) \right| v(\vec{p}, \vec{\varepsilon}) \right\rangle = \varepsilon^i e^{-i\vec{p} \cdot \vec{x}} \quad (2.18)$$

where ε^i transforms as

$$\varepsilon^i \rightarrow R_{ij} \varepsilon^j \quad (2.19)$$

Then the particle carries a real three vector that rotates the state $|v(\vec{p}, \vec{\varepsilon})\rangle$. This vector is the *polarisation vector* of the state.

We can construct fields corresponding to any spin j representation of the rotation group. Let $R_{ab}^{(j)}$ be a rotation matrix in the spin j representation. This can be notated by a

$(2j + 1) \times (2j + 1)$ matrix.

A spin j field would transform under rotations as

$$W^a(\vec{x}) \rightarrow W'^a(\vec{x}) = R_{ab}^j W^b(R^{-1}\vec{x}) \quad (2.20)$$

The field $W^a(\vec{x})$ would destroy a particle according to the matrix element

$$\left\langle 0 \left| W^a(\vec{x}) \right| W(\vec{p}, \vec{\eta}) \right\rangle = \eta^a e^{-i\vec{p} \cdot \vec{x}} \quad (2.21)$$

The particle w could carry a $(2j + 1)$ -dimensional polarisation vector η^a , which would transform a spin j vector under rotation.

Example 2.3.2 (). For $j = \frac{1}{2}$, η would be a 2 component spinor.

The matrix elements (2.18) and (2.21) aren't relativistically invariant.

Example 2.3.3 (Spin 1 Matrix Elements). We generalise (2.18) as $i \rightarrow \mu$

$$\left\langle 0 \left| V^\mu(x^\mu) \right| v(p^\mu, \varepsilon^\mu) \right\rangle = \varepsilon^\mu e^{-ip_\mu x^\mu} \quad (2.22)$$

where ε^μ is promoted to a 4-vector.

To preserve Lorentz invariance, the norm of the spin 1 particle should be proportional to $-\varepsilon^\mu \varepsilon_\mu$. We have a negative sign since the expression should be positive if ε is a spacelike unit vector. ε^μ may also be a timelike vector, e.g. $\varepsilon^\mu = (1, 0, 0, 0)$.

Should the particle have a negative norm, then that would correspond to a negative probability. Consequently, the spin 1 wave equation must forbid a timelike ε^μ appearing.

Photons are spin 1 particles, thus their wave equation solves this problem. Maxwell's equations in terms of $A_\mu(\vec{x})$ have propagating solutions with only two possible polarisation vectors

$$\vec{A}(\vec{x}) = \vec{\varepsilon} e^{-iEt + i\vec{p} \cdot \vec{x}} \quad (2.23)$$

where $E = |p|$ (massless) and

$$\vec{\varepsilon} \cdot \vec{p} = 0 \quad (2.24)$$

We can view (2.23) as the wavefunction of the photon. Under a Lorentz rotation, rotation (2.23) is unchanged. However, under a Lorentz boost, the exponential is unchanged, but the polarisation vector acquires a component in the time direction. The Lorentz transformation of $\vec{\varepsilon}$ does not affect the photon's interactions as long as the electromagnetic current is conserved. We obtain a description of the states that are Lorentz invariant, though not manifestly so. Thus, we can quantise the Maxwell field using only states of positive norm.

We can't simply generalise to spin 1 as we need the Higg's mechanism for this.

Consider the variational principle for the Maxwell field:

$$A^\mu(x^\mu) = \left(-\phi(x^\mu), -\vec{A}(x^\mu) \right)^\mu \quad (2.25)$$

where ϕ and $\vec{A}$ are scalar and vector potentials respectively

$$F^{\mu\nu} := \partial^\mu A^\nu - \partial^\nu A^\mu \quad (2.26)$$

varying out tensor compoents, $\partial^\mu = \left(\frac{\partial}{\partial t}, -\vec{\nabla}\right)^\mu$.

$$\begin{aligned} F^{0i} &= -\nabla^i \phi - \partial_0 A^i = E^i \\ F^{ij} &= \nabla^i A^j - \nabla^j A^i = \varepsilon^{ijk} B^k. \end{aligned}$$

The Maxwell Lagrangian density is

$$\mathcal{L}_{MW} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + J^\mu A_\mu \quad (2.27)$$

where we include a new field $J^\mu(x^\mu)$ to represent an external electromagnetic current, which satisfies current conservation

$$J^\mu = \left(\rho, \vec{J}\right)^\mu \quad (2.28)$$

Explicitly,

$$0 = \partial_\mu J^\mu = \partial_t \rho + \vec{\nabla} \cdot \vec{J} \quad (2.29)$$

If we vary (2.27) with respect to $A^\mu(x^\mu)$, the variation is

$$\partial \mathcal{L}_{MW} = -\frac{1}{2} (\partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu) F^{\mu\nu} + \delta A_\nu J^\nu \quad (2.30)$$

Since, $F^{\mu\nu} = -F^{\nu\mu}$, then

$$\begin{aligned} \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu &= 2\partial_\mu A_\nu \\ -\frac{1}{2} (\partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu) &= -\partial_\mu \delta A_\nu \\ \delta S &= \int d^4x [-(\partial_\mu \delta A_\nu) F^{\mu\nu} + \delta A_\nu J^\nu] \\ \partial_\mu (\delta A_\nu F^{\mu\nu}) &= \delta A_\nu \partial_\mu F^{\mu\nu} + (\partial_\mu \delta A_\nu) F^{\mu\nu} \\ \delta S &= \int d^4x [\delta A_\nu \partial_\mu F^{\mu\nu} - \partial_\mu (\delta A_\nu F^{\mu\nu}) + \delta A_\nu J^\nu]. \end{aligned}$$

The second term is a boundary term

$$-\int d^4x \partial_\mu (\delta A_\nu F^{\mu\nu}) = -\int_{\partial\mathcal{M}} d\Sigma_\mu \delta A_\nu F^{\mu\nu}.$$

which vanishes as we require that $\delta A_\nu = 0$ at infinity. Thus we are left with

$$\begin{aligned} \delta S &= \int d^4x \delta A_\nu [\partial_\mu F^{\mu\nu} + J^\nu] \\ \implies \partial_\mu F^{\mu\nu} &= -J^\nu. \end{aligned}$$

After substitution we find that

$$\begin{aligned} \nabla^i E^i &= \rho \\ \varepsilon^{ijk} \nabla^j B^k - \partial_t E^i &= J^i. \end{aligned}$$

which are the natural unit, in-homogeneous Maxwell equations.

2.4 The Dirac Field

The Dirac equation aims to describe spin $\frac{1}{2}$ particles e.g. the electron. To do this we consider the Dirac matrices

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \quad (2.31)$$

It is a fact that the smallest representation of these matrices are for 4×4 matrices. One such representation is

$$\gamma^0 = \begin{bmatrix} \mathbf{1} & 0 \\ 0 & \mathbf{1} \end{bmatrix} \quad \gamma^i = \begin{bmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{bmatrix} \quad (2.32)$$

where the elements of these matrices are 2×2 blocks and σ^i are the 2×2 Pauli matrices. The Pauli matrices satisfy

$$\sigma^i \sigma^j = \delta^{ij} \mathbf{1} + i\epsilon^{ijk} \sigma_k \quad (2.33)$$

Using the Dirac field we can write the Dirac equation

$$(i\gamma^\mu \partial_\mu - m) \psi = 0 \quad (2.34)$$

Proposition 2.4.1 (Properties of the Dirac Equation).

1. *For all solutions to the Dirac equation that solution is also a solution the Klein–Gordon equation This implies that the exponential part at least takes the form of $\psi \sim e^{i\vec{p} \cdot \vec{x}}$ and that the equation is Lorentz invariant.*
2. *The solution to the Dirac equation for $\vec{p} = 0$ has exactly two positive and two negative solutions*
3. *A field satisfying the Dirac equation has a vector current which is conserved For electrons this operator would be the electric charge current.*

To fully prove that the Dirac equation is relativistic, we would need to define the Lorentz transformation properties of γ^μ carefully and prove that these matrices transform as a 4-vector¹.

Properties of Dirac Equation.

1. To prove the first property of the Dirac equation multiply by $(i\gamma^\mu \partial_\mu + m)$

$$\begin{aligned} (i\gamma^\nu \partial_\nu + m) \cdot (i\gamma^\mu \partial_\mu - m) \psi &= 0 \\ (-\gamma^\nu \gamma^\mu \partial_\mu \partial_\nu - m^2) \psi &= 0 \end{aligned}$$

□

¹See Peskin QFT